

=== CONFIGURACION GNB (1779214868) ===

```
# gnb.yaml
# This example configuration outlines how to configure the srsRAN Project gNB to create a single FDD cell
# transmitting in band 3, with 10 MHz bandwidth and 15 kHz sub-carrier-spacing. A USRP B200 is configured
# as the RF frontend using split 8, with a Leo Bodnar GPSDO as an external reference clock.

# This particular configuration is intended to be used with srsUE. Note the `pdcch` parameters set in the
# `cell_cfg` section. These are set to match the capabilities of srsUE.

cu_cp:
  amf:
    addr: 10.53.1.2
    port: 38412
    bind_addr: 10.53.1.3
    supported_tracking_areas:
      - tac: 7
    plmn_list:
      - plmn: "20893"
    tai_slice_support_list:
      - sst: 1 # SST 1: eMBB (enhanced Mobile Broadband) - Ref. 3GPP TS 23.501 Table 5.15.2.2-1

ru_sdr:
  device_driver: uhd
  device_args: type=b200
  clock: external
  srate: 15.36 # Sample rate for 10 MHz bandwidth
  tx_gain: 80
  rx_gain: 40

cell_cfg:
  dl_arfcn: 368500 # Example ARFCN, adjust for actual band 3 frequency
  band: 3
  channel_bandwidth_MHz: 10 # Changed to 10 MHz as requested
  common_scs: 15
  plmn: "20893"
  tac: 7
  pdcch:
    dedicated:
      ss2_type: common
      dci_format_0_1_and_1_1: false
    common:
      ss0_index: 0
      coresets0_index: 6 # Changed to 6 for 10 MHz bandwidth
  prach:
    prach_config_index: 1
  pdsch:
    mcs_table: qam64
  pusch:
    mcs_table: qam64

log:
  filename: /tmp/gnb.log
  all_level: info

pcap:
  mac_enable: enable
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: enable
  ngap_filename: /tmp/gnb_ngap.pcap
```

=== CONFIGURACION UE (1779214868) ===

```
# ue.conf
# srsUE configuration file

[rrc]
# RRC configuration
# The following parameters are mandatory for srsUE
# and must match the gNB configuration.
mcc = 208
mnc = 93
imsi = 208930000000001 # Example IMSI, ensure it's registered in Open5GS
key = c9e8763286b5b9ff2177569700000000 # Example test key
opc = 981ca10029ee5405d06e6a865063e256 # Example test OPC
apn = internet

[nas]
# NAS configuration
# The following parameters are mandatory for srsUE
# and must match the gNB configuration.
# sst = 1 # Slice/Service Type (eMBB)
# sd = 1 # Slice Differentiator (optional)

[usim]
# USIM configuration
# The following parameters are mandatory for srsUE
# and must match the gNB configuration.
# k = 00112233445566778899AABBCCDDEEFF # Example K value (if not using key/opc)

[rf]
# RF configuration
# The following parameters are mandatory for srsUE
# and must match the gNB configuration.
device_name = softmodem # Use softmodem for ZMQ connection
gnb_addr = 10.53.1.3 # gNB IP address
gnb_port = 2000 # gNB ZMQ port (gNB listens, UE connects)

[log]
# Logging configuration
filename = /tmp/ue.log
all_level = info

[pcap]
# PCAP logging configuration
mac_enable = enable
mac_filename = /tmp/ue_mac.pcap
```

```
# docker-compose.yml
version: '3.8'

networks:
  open5gs-net:
    ipam:
      config:
        - subnet: 10.53.1.0/24

services:
  mongodb:
    container_name: mongodb
    image: mongo:4.4
    networks:
      open5gs-net:
        ipv4_address: 10.53.1.100
    volumes:
      - ./data/db:/data/db
    restart: always

  open5gs-amf:
    container_name: open5gs-amf
    image: open5gs/open5gs:latest
    command: open5gs-amfd
    volumes:
      - ./open5gs-config:/open5gs/config
    networks:
      open5gs-net:
        ipv4_address: 10.53.1.2
    depends_on:
      - mongodb
      - open5gs-nrf
    restart: always

  open5gs-smf:
    container_name: open5gs-smf
    image: open5gs/open5gs:latest
    command: open5gs-smfd
    volumes:
      - ./open5gs-config:/open5gs/config
    networks:
      open5gs-net:
        ipv4_address: 10.53.1.4
    depends_on:
      - mongodb
      - open5gs-nrf
    restart: always

  open5gs-upf:
    container_name: open5gs-upf
    image: open5gs/open5gs:latest
    command: open5gs-upfd
    volumes:
      - ./open5gs-config:/open5gs/config
    networks:
      open5gs-net:
        ipv4_address: 10.53.1.7
    depends_on:
      - mongodb
      - open5gs-smf
    cap_add:
      - NET_ADMIN
    privileged: true
    restart: always
```

open5gs-nrf:
 container_name: open5gs-nrf
 image: open5gs/open5gs:latest
 command: open5gs-nrfd
 volumes:
 - ./open5gs-config:/open5gs/config
 networks:
 open5gs-net:
 ipv4_address: 10.53.1.10
 depends_on:
 - mongodb
 restart: always

open5gs-udm:
 container_name: open5gs-udm
 image: open5gs/open5gs:latest
 command: open5gs-udmd
 volumes:
 - ./open5gs-config:/open5gs/config
 networks:
 open5gs-net:
 ipv4_address: 10.53.1.12
 depends_on:
 - mongodb
 - open5gs-nrf
 restart: always

open5gs-ausf:
 container_name: open5gs-ausf
 image: open5gs/open5gs:latest
 command: open5gs-ausfd
 volumes:
 - ./open5gs-config:/open5gs/config
 networks:
 open5gs-net:
 ipv4_address: 10.53.1.11
 depends_on:
 - mongodb
 - open5gs-nrf
 restart: always

open5gs-pcf:
 container_name: open5gs-pcf
 image: open5gs/open5gs:latest
 command: open5gs-pcfd
 volumes:
 - ./open5gs-config:/open5gs/config
 networks:
 open5gs-net:
 ipv4_address: 10.53.1.13
 depends_on:
 - mongodb
 - open5gs-nrf
 restart: always

open5gs-nssf:
 container_name: open5gs-nssf
 image: open5gs/open5gs:latest
 command: open5gs-nssfd
 volumes:
 - ./open5gs-config:/open5gs/config
 networks:
 open5gs-net:
 ipv4_address: 10.53.1.14
 depends_on:
 - mongodb
 - open5gs-nrf

```

restart: always

open5gs-scp:
  container_name: open5gs-scp
  image: open5gs/open5gs:latest
  command: open5gs-scpd
  volumes:
    - ./open5gs-config:/open5gs/config
  networks:
    open5gs-net:
      ipv4_address: 10.53.1.200
  depends_on:
    - mongodb
    - open5gs-nrf
  restart: always

# Open5GS Core Configuration (open5gs-config/open5gs.yaml)
# This file should be placed in a directory named 'open5gs-config'
# relative to your docker-compose.yml.
#
# db_uri: mongodb://mongodb/open5gs
#
# logger:
#
# global:
#   parameter:
#     no_nrf: false # NRF is enabled
#     no_scp: false # SCP is enabled
#     no_sepp: true # SEPP is disabled for simplicity
#     no_amf: false # AMF is enabled
#     no_smf: false # SMF is enabled
#     no_upf: false # UPF is enabled
#     no_auks: false # AUSF is enabled
#     no_udm: false # UDM is enabled
#     no_pcf: false # PCF is enabled
#     no_nssf: false # NSSF is enabled
#     no_bsf: false # BSF is enabled
#     no_uds: false # UDR is enabled
#     no_mme: true # MME is disabled (5GC)
#     no_sgw_c: true # SGW-C is disabled (5GC)
#     no_sgw_u: true # SGW-U is disabled (5GC)
#     no_pcrf: true # PCRF is disabled (5GC)
#     no_hss: true # HSS is disabled (5GC)
#
# amf:
#   sbi:
#     server:
#       - address: 10.53.1.2
#         port: 7777
#     client:
#       scp:
#         - uri: http://10.53.1.200:7777
#   ngap:
#     server:
#       - address: 10.53.1.2
#   metrics:
#     server:
#       - address: 10.53.1.2
#         port: 9090
#   access_control:
#     - plmn_id:
#         mcc: 208
#         mnc: 93
#   guami:
#     - plmn_id:
#         mcc: 208
#         mnc: 93
#     amf_id:

```

```
#         region: 2
#         set: 1
# tai:
#   - plmn_id:
#       mcc: 208
#       mnc: 93
#   tac: 7 # Matches gNB TAC
# plmn_support:
#   - plmn_id:
#       mcc: 208
#       mnc: 93
#   s_nssai:
#       - sst: 1 # SST 1: eMBB
# security:
#   integrity_order : [ NIA2, NIA1, NIA0 ]
#   ciphering_order : [ NEA0, NEA1, NEA2 ]
# network_name:
#   full: Open5GS
# amf_name: open5gs-amf0
# time:
#   t3512:
#       value: 540      # 9 minutes * 60 = 540 seconds
#
# smf:
#   sbi:
#     server:
#       - address: 10.53.1.4
#       port: 7777
#     client:
#     scp:
#       - uri: http://10.53.1.200:7777
#   pfcp:
#     server:
#       - address: 10.53.1.4
#     client:
#     upf:
#       - address: 10.53.1.7
#   metrics:
#     server:
#       - address: 10.53.1.4
#       port: 9090
#   session:
#     - subnet: 10.45.0.0/16
#     gateway: 10.45.0.1
#     - subnet: 2001:db8:cafe::/48
#     gateway: 2001:db8:cafe::1
#   dns:
#     - 8.8.8.8
#     - 8.8.4.4
#     - 2001:4860:4860::8888
#     - 2001:4860:4860::8844
#   mtu: 1400
#
# upf:
#   pfcp:
#     server:
#       - address: 10.53.1.7
#   gtpu:
#     server:
#       - address: 10.53.1.7
#   session:
#     - subnet: 10.45.0.0/16
#     gateway: 10.45.0.1
#     dev: ogstun
#     - subnet: 2001:db8:cafe::/48
#     gateway: 2001:db8:cafe::1
#     dev: ogstun
#   metrics:
```

```
# server:
#   - address: 10.53.1.7
#     port: 9090
#
# nrf:
#   serving:
#     - plmn_id:
#         mcc: 208
#         mnc: 93
#   sbi:
#     server:
#       - address: 10.53.1.10
#         port: 7777
#
# scp:
#   sbi:
#     server:
#       - address: 10.53.1.200
#         port: 7777
#     client:
#       nrf:
#         - uri: http://10.53.1.10:7777
#
# ausf:
#   sbi:
#     server:
#       - address: 10.53.1.11
#         port: 7777
#     client:
#       scp:
#         - uri: http://10.53.1.200:7777
#
# udm:
#   hnet:
#     - id: 1
#       scheme: 1
#       key: @build_configs_dir@/open5gs/hnet/curve25519-1.key
#     - id: 2
#       scheme: 2
#       key: @build_configs_dir@/open5gs/hnet/secp256r1-2.key
#   sbi:
#     server:
#       - address: 10.53.1.12
#         port: 7777
#     client:
#       scp:
#         - uri: http://10.53.1.200:7777
#
# pcf:
#   sbi:
#     server:
#       - address: 10.53.1.13
#         port: 7777
#     client:
#       scp:
#         - uri: http://10.53.1.200:7777
#
# metrics:
#   server:
#     - address: 10.53.1.13
#       port: 9090
#
# policy:
#   - supi_range:
#       - 208930000000001-208930999999999 # Matches example IMSI
#   slice:
#     - sst: 1 # SST 1: eMBB
#       default_indicator: true
#       session:
#         - name: internet
```

```
#         type: 3 # 1:IPv4, 2:IPv6, 3:IPv4v6
#         ambr:
#             downlink:
#                 value: 1
#                 unit: 3 # 0:bps, 1:Kbps, 2:Mbps, 3:Gbps, 4:Tbps
#             uplink:
#                 value: 1
#                 unit: 3
#         qos:
#             index: 9 # Default QCI for internet
#         arp:
#             priority_level: 8
#             pre_emption_vulnerability: 1
#             pre_emption_capability: 1
#
# nssf:
#     sbi:
#         server:
#             - address: 10.53.1.14
#               port: 7777
#         client:
#             scp:
#                 - uri: http://10.53.1.200:7777
#             nsi:
#                 - uri: http://10.53.1.10:7777
#             s_nssai:
#                 sst: 1
#
# bsf:
#     sbi:
#         server:
#             - address: 10.53.1.15
#               port: 7777
#         client:
#             scp:
#                 - uri: http://10.53.1.200:7777
#
# udr:
#     sbi:
#         server:
#             - address: 10.53.1.20
#               port: 7777
#         client:
#             scp:
#                 - uri: http://10.53.1.200:7777
```